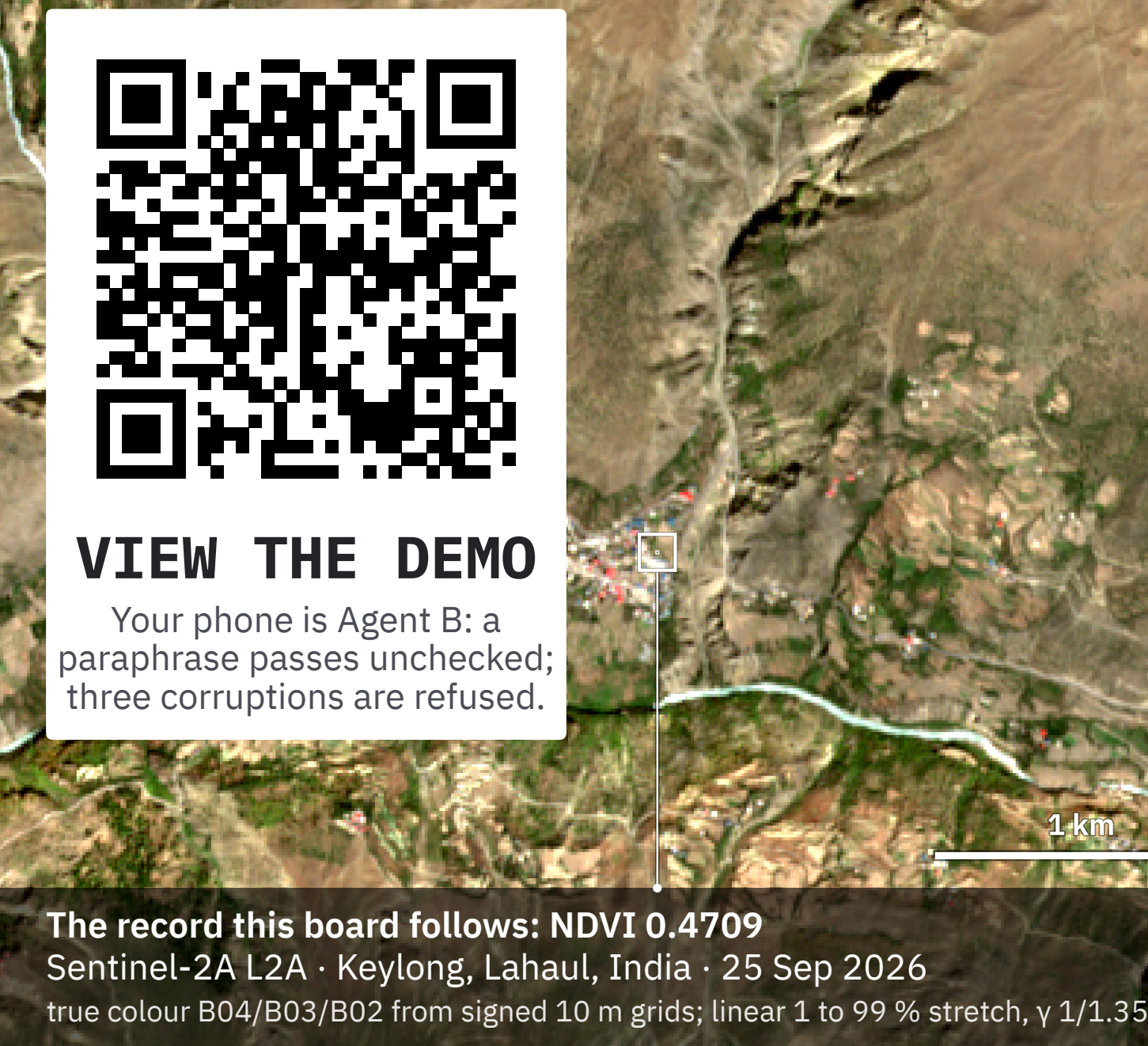


EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol for AI Agents over Foundation-Model Embeddings

Agents hand each other evidence references, not paraphrases.

The receiver resolves the reference, re-hashes the record, re-reads the source. A signature fixes the record's bytes, not the measurement.

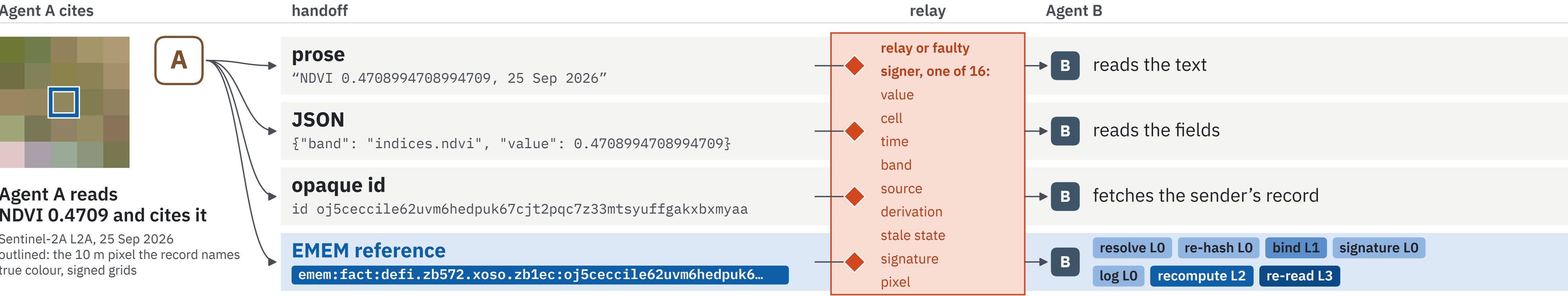
Jaya Kumari · Avijeet Singh · Vortex AI · avijeet@vortex.ai
Agentic AI for Earth Observation · BIFOLD and ESA Φ-lab · Berlin · Poster Session 1 · 19 Oct 2026 · emem.dev · github.com/Vortex-AI/emem (Apache-2.0)



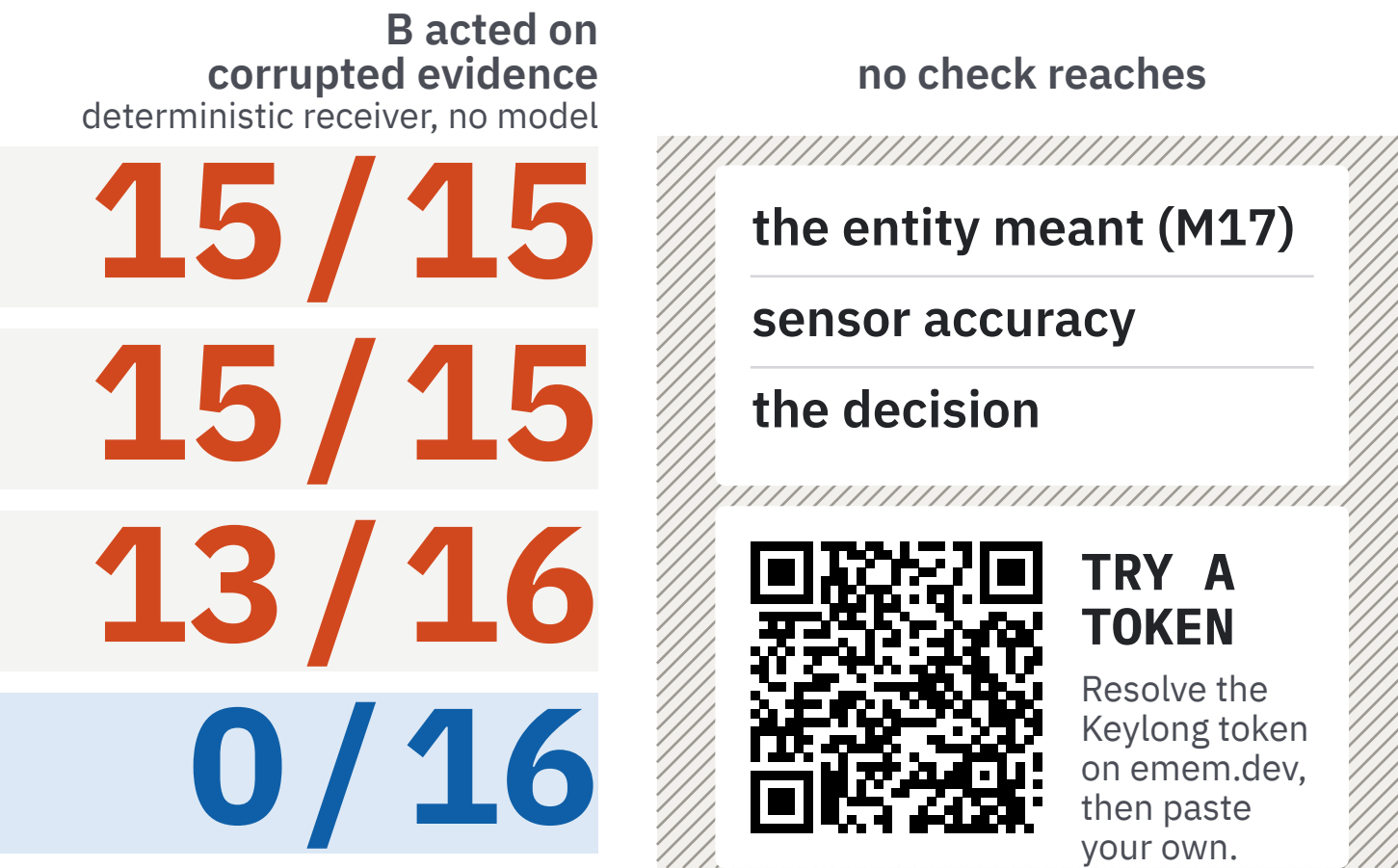
1 What survives an agent handoff?

Handed prose, B acted on 15 of 15 corruptions; handed a reference it checks, on 0 of 16.

A relay or a faulty signer corrupts one signed NDVI record in one of 16 ways; Agent B resolves, re-hashes, binds, checks, recomputes and re-reads.



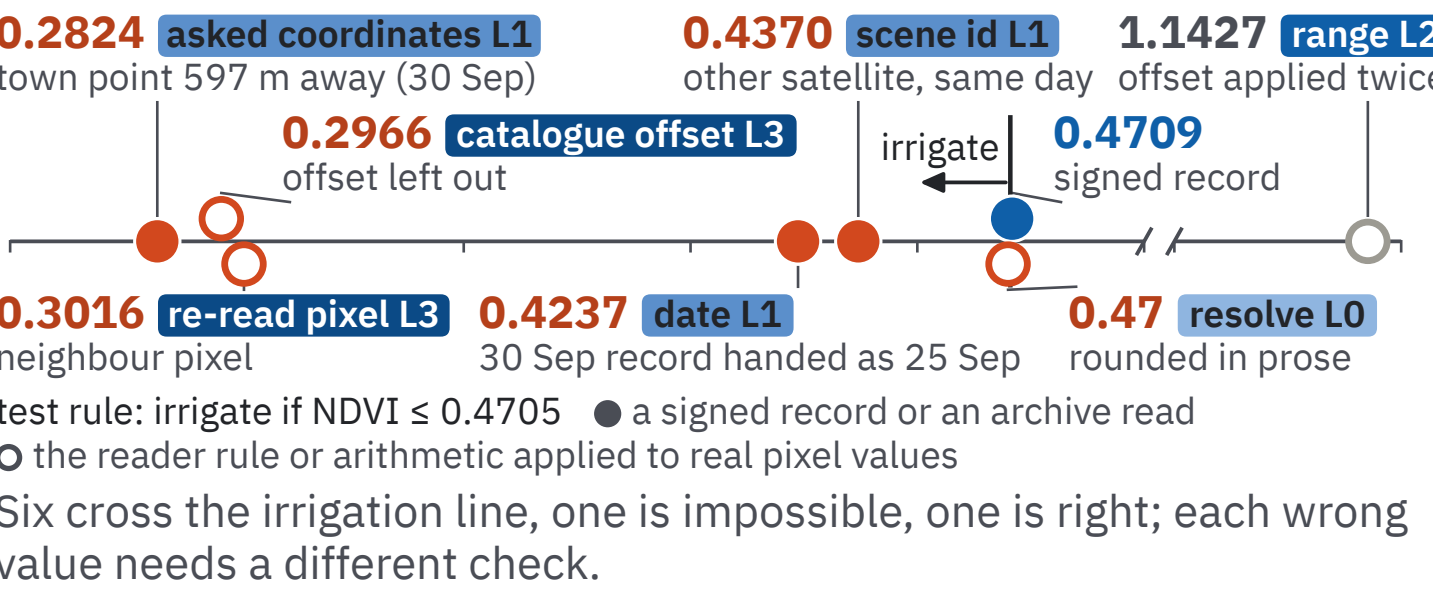
With a model as B and a forged cell, Claude Haiku 4.5 declined 5 of 5; Qwen2.5-3B dropped the cell from all 24 calls and acted 5 of 5: a reference protects only a receiver that keeps it whole and obeys a refusal.



2 What can one NDVI become between agents?

Eight answers; one is right

One question, NDVI at the Keylong field on 25 Sep 2026; real records, archive pixels and handling rules give eight values.



3 Why does an agent handoff need a record?

Our own errors name the checks

Each error happened inside emem, a system built for evidence, or in our tests; each has a published precedent.

errors may propagate silently across steps" (Munir et al. 2026, arXiv 2604.24919) each rung happened, in emem or in our tests	none · L4: fixed 30 Sep 2026
8 Wrong thing "this image" resolved to a hair salon in Ontario receipt, Merkle proof and state chain all valid toponym ambiguity (Gritta et al. 2018)	re-read only · L3: fixed 28 Sep 2026
7 Wrong pixel, signed the reader rounded the pixel index 162 of 200 sampled records GDAL RFC 33: half-pixel shift; one-pixel misregistration: error > 50 % of NDVI differences (Townshend 1992)	re-read · L3: fixed
6 Missing data signed as 0 off-tile pixels signed as 0 a forest-loss screen passes Hansen lossyear 0 means no loss	re-read · L3: fixed
5 Wrong scale or offset WorldPop signed per pixel, not per km²: 1.77x low one pixel: 0.4709, 0.2966, 1.1427 under three offset rules Element84 items say "offset applied" and "apply -0.1"	recompute · L2: fixed catalogue · L3: fixed
4 Wrong version 918.0 m, then 915.07 m, one band name an earlier citation looks wrong GFC2020 V3 cut forest cover by more than 20 % in the Cerrado	source + as-of · L1: both kept
3 Wrong place coordinates given, town point answered 597 m; NDVI 0.28 for the field's 0.42 GDAL 3 follows CRS axis order (RFC 73)	asked coordinates · L1: fix unconfirmed
2 Wrong date asked 23 Sep, served 25 Sep 10 of 10 agents saw one scene twice; 3 said no 23 Sep scene existed STAC item search defines no default order	bind the date · L4: open
1 Lost in the handoff 0.47 for 0.4709 5 of 5 receivers irrigated Perez et al., ICLR 2025: LLM transmission chains drift	resolve the reference · L0: test rule

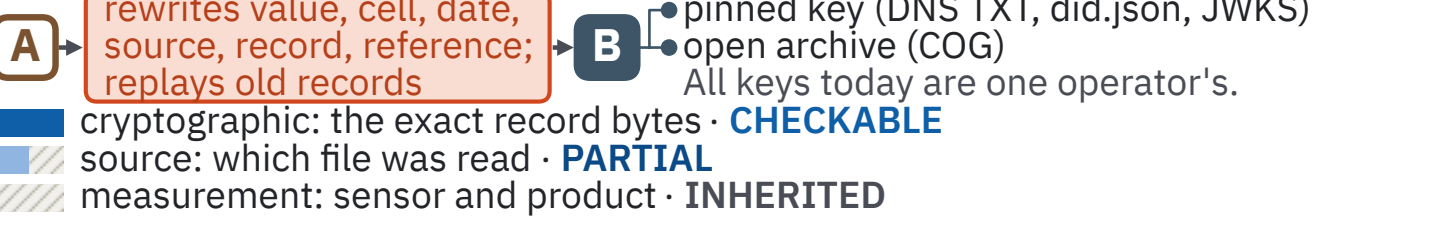
Signatures prevented none; the record made each checkable at a named layer, except the top row.
A check protects only if it runs: emem's resolver once reported a cell match it never tested; one framework adapter returns a refusal as plain text; MAST: no or incomplete verification (FM-3.2).

We ask four questions

- RQ1 Can a receiver detect corruption without trusting the sender? → 1, 5
RQ2 Which corruption classes can it detect? → 5
RQ3 Does a reference beat prose, JSON or an opaque id? → 1
RQ4 What stays unverifiable with the reference intact? → 6, 7
H1 A checked reference refuses corruptions that keep the stated value.
H2 Only a source re-read catches a signer's wrong pixel. H3 A historical reference keeps the cited state.

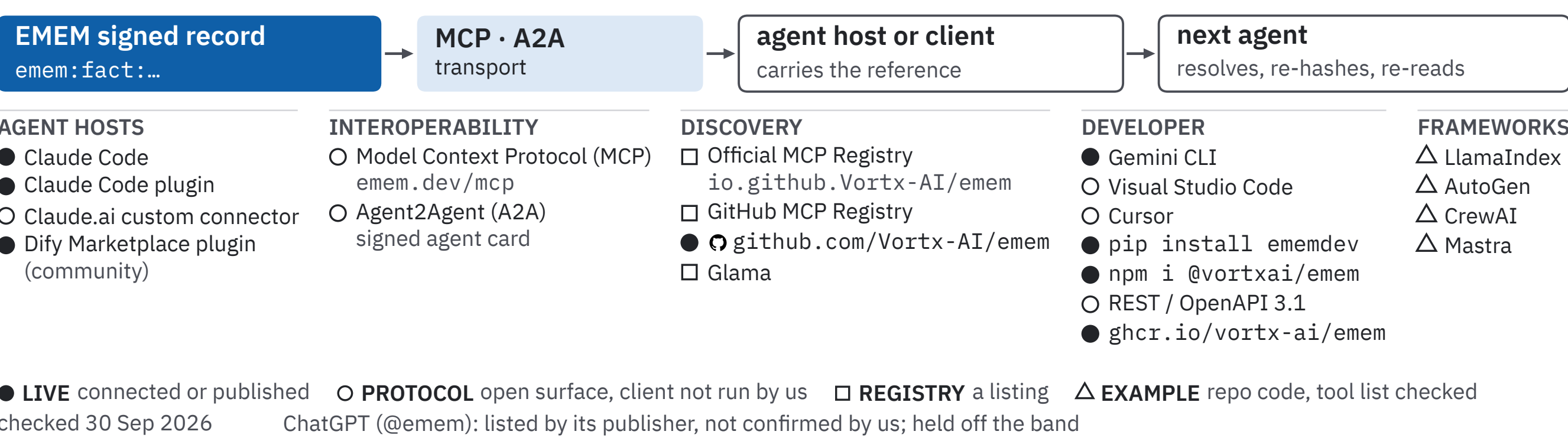
A relay can rewrite, not sign

A relay can rewrite what it carries but cannot sign under the pinned key or match an address; a signer's misread pixel is caught only by a re-read. Out of scope: a compromised key, a wrong sensor, the entity meant, the decision.



12 One evidence protocol, multiple agent runtimes.

One reference resolved to one address and one value through 11 client paths, with receipt signatures checked on 9 (30 Sep 2026).
A listing is not an integration. Not run by us: ChatGPT, claude.ai, Diffy, VS Code, Cursor.

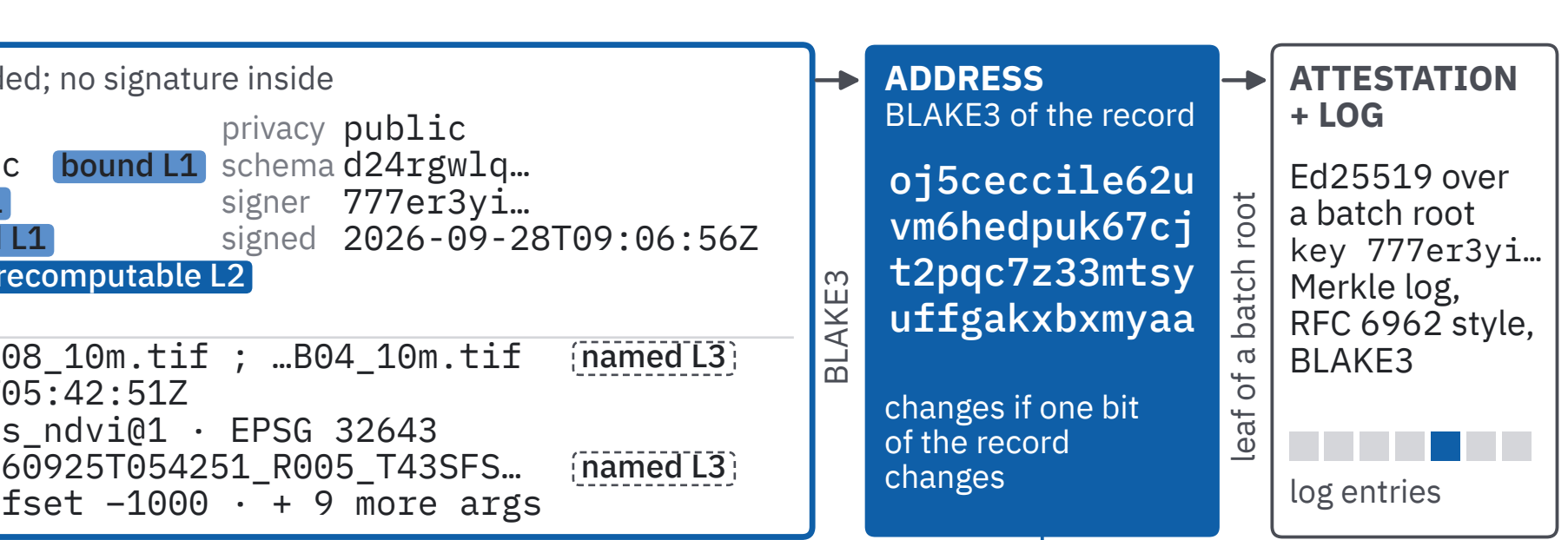


4 What exactly is handed over?

The address names the record, not the pixel.

The receiver gets 84 characters; everything else it fetches and checks.

INSPECT THE RECORD The 1,115 bytes behind the main example, decoded.



The address is the BLAKE3 hash of the 1,115-byte record, which names its source files and the point it read but holds no hash of them; a batch signature covers such addresses. Re-signing makes a new record: one Bengaluru value has seven.

5 Which check stops which corruption?

Each of five checks alone stops a corruption.

